Network Wave Equation: soft nodes, nonlinear periodic solutions and graph characterizations

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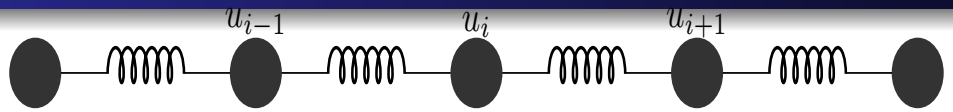
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Flow of a miscible quantity on a network

- gas, water or power distribution networks. a simplified version of road traffic, flow of nerve impulse in the brain.
- static aspect: long been studied, operations research
- generalize conservation laws
  $u_t + \partial_x q = 0,$
- Dynamic problem: solved numerically
  In general, difficult to analyze
  classes of systems which can be analyzed (dissipation at nodes)

Discrete dynamical systems



Spring-mass chain.

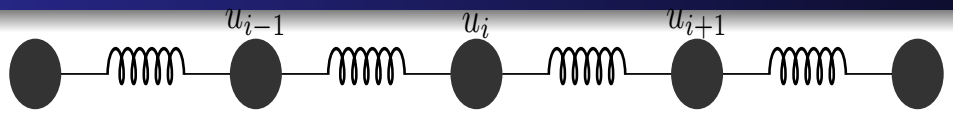
Newton's second law for coupled harmonic oscillators

$$m\frac{\mathrm{d}^2 u_i}{\mathrm{d}t^2} = k\left(u_{i-1} - u_i\right) + k\left(u_{i+1} - u_i\right). \tag{1}$$

Finite difference discretization of the continuous wave equation

$$\frac{\mathrm{d}^2 u}{\mathrm{d}t^2} = \Delta u. \tag{2}$$

Discrete dynamical systems



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Coupled nonlinear differential equations

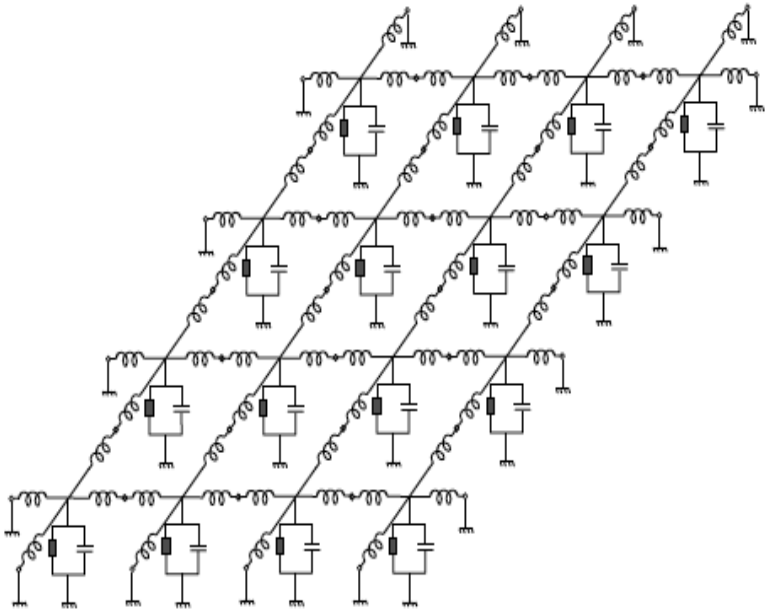
- Intersite nonlinearity: **the Fermi-Pasta-Ulam-Tsingou model**

$$\frac{\mathrm{d}^2 u_i}{\mathrm{d}t^2} = u_{i-1} - 2u_i + u_{i+1} - \left(u_i - u_{i-1}\right)^3 - \left(u_i - u_{i+1}\right)^3. \tag{3}$$

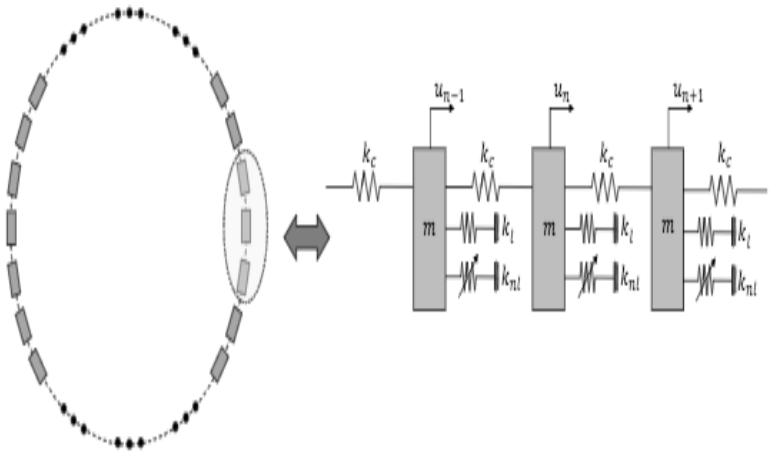
- On-site nonlinearity: **the discrete Φ^4 model**

$$\frac{\mathrm{d}^2 u_i}{\mathrm{d}t^2} = u_{i-1} - 2u_i + u_{i+1} - u_i^3. \tag{4}$$

Nonlinear lattices



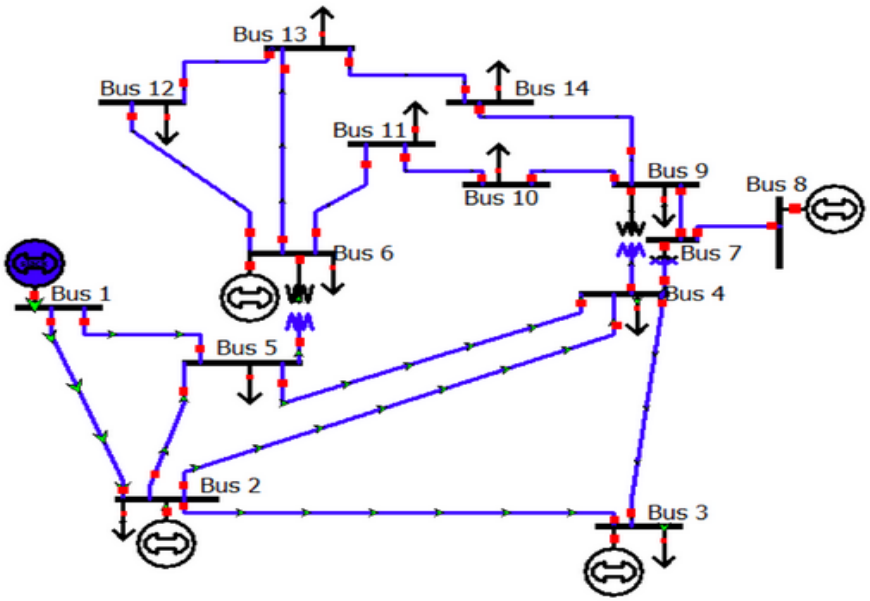
A 4×4 array of the unit cells of a two-dimensional tunnel diode or Josephson junction array (**Scott, 2003**).



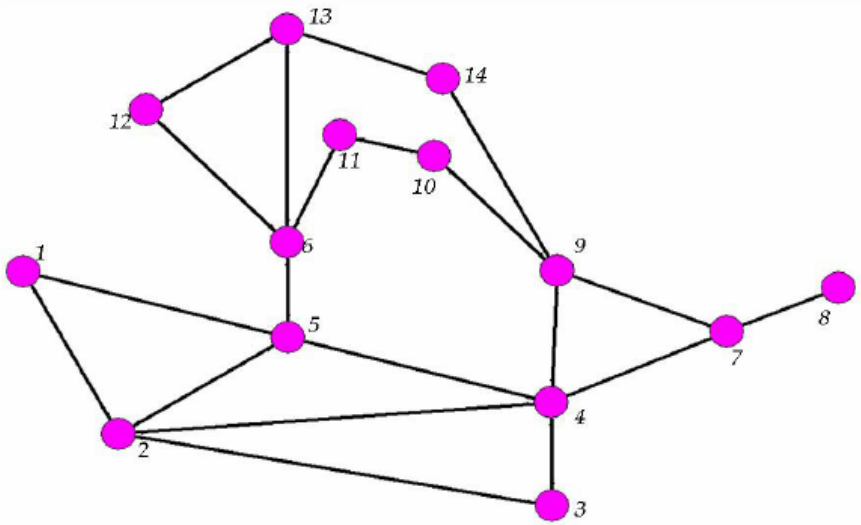
Mechanical cyclic structure (**Fontanela, Grolet, Salles, Chabchoub, Hoffmann, J. Sound and Vibration 413, 2018**).

Complex systems

Network (graph) = set of nodes (vertices) joined by edges (branches).



IEEE 14-Bus System: a simple approximation of the American Electric Power system (1962). It has 14 buses, 5 generators, and 11 loads.



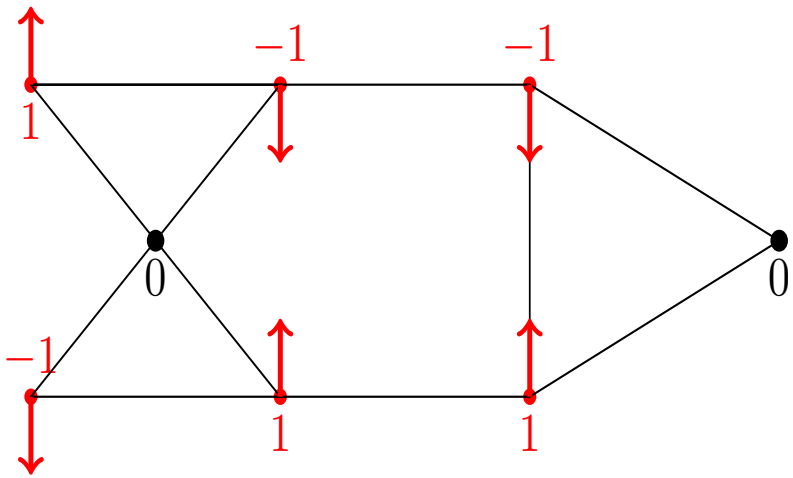
The schematic view of IEEE 14-Bus System.

Wave equation on general networks

Wave equation on general graphs

$$\frac{\mathrm{d}^2 u}{\mathrm{d}t^2} = -\underbrace{\Delta}_{\text{Graph Laplacian}} u. \qquad (5)$$

Normal modes $u = z\mathrm{e}^{\mathrm{i}\omega t}$.
Eigenvalue problem $\Delta z = \omega^2 z$.

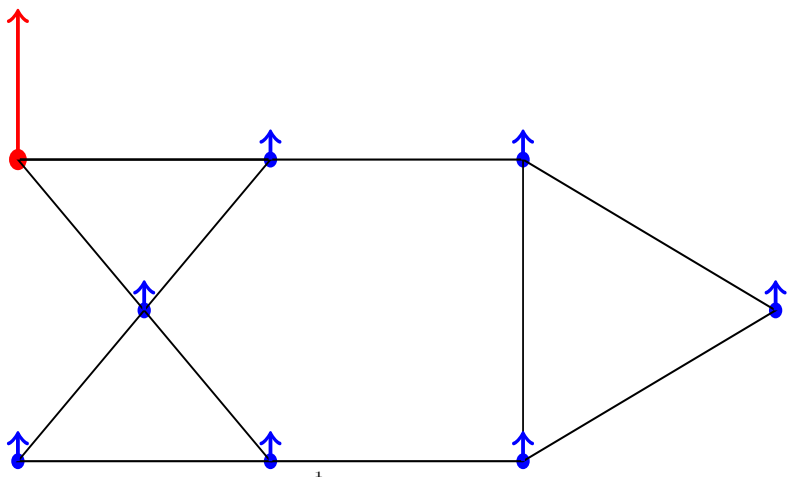


An illustration of nonlinear normal modes.

Nonlinear graph wave equation

$$\frac{\mathrm{d}^2 u}{\mathrm{d}t^2} = -\Delta u - u^3. \qquad (6)$$

- Nonlinear periodic solutions:
  - linear coupling $\rightarrow$ **normal modes**.
  - nonlinearity $\rightarrow$ **localized solutions**.
- Interplay between nonlinear dynamics and graph spectra.



An illustration of localized solutions.

Outline

1. Network wave equation

Plan

1. Network wave equation

Discrete Gradient

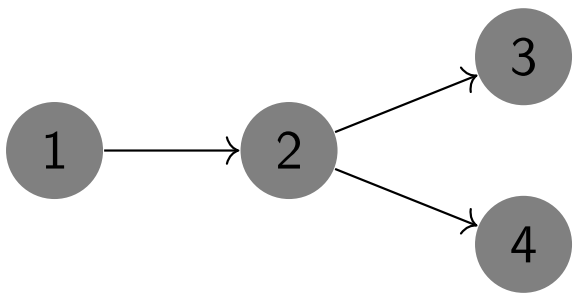
A graph $\mathcal{G}(\mathcal{V}, \mathcal{E})$ with vertex set $\mathcal{V}$ of cardinality N and edge set $\mathcal{E}$ of cardinality M.

Incidence matrix $Q \in \mathcal{M}_{N,M}(\mathbb{Z})$

$$Q_{je} = \begin{cases} -1 & \text{if branch } e \text{ starts from node } j, \\ 1 & \text{if branch } e \text{ finishes at node } j, \\ 0 & \text{otherwise.} \end{cases} \tag{7}$$

Gradient of graph : $\nabla = Q^T$.

$$\nabla f = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{pmatrix} \begin{pmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{pmatrix} = \begin{pmatrix} f_2 - f_1 \\ f_3 - f_2 \\ f_4 - f_2 \end{pmatrix}$$



Paw graph.

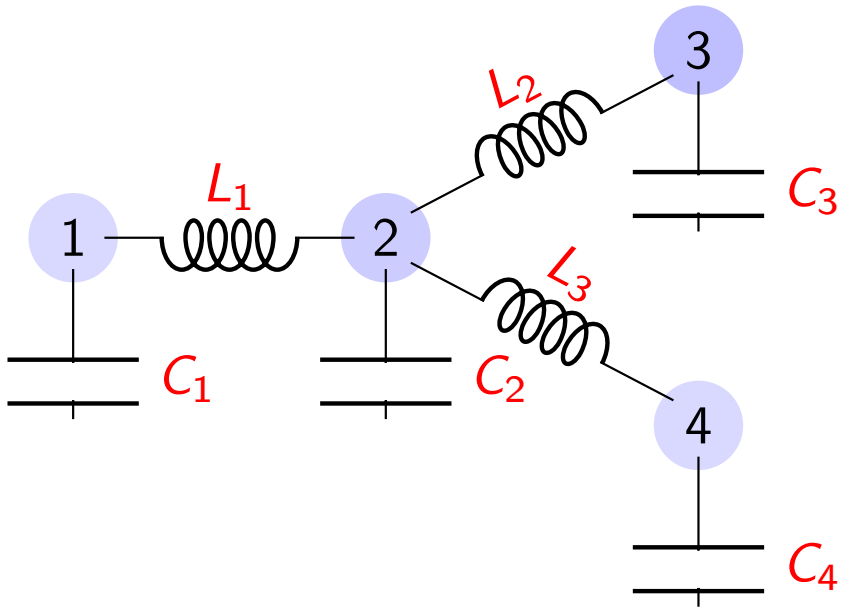
Discrete conservation laws

Generalize conservation laws $\frac{\partial u}{\partial t} + \frac{\partial q}{\partial x} = 0$.

Array of inductances and capacities

$$C\frac{\mathrm{d}v}{\mathrm{d}t} + \nabla^T i = s, \tag{8}$$

$$L\frac{\mathrm{d}i}{\mathrm{d}t} - \nabla v = 0. \tag{9}$$



Electrical Network.

$v = (v_1,\ v_2,\ v_3,\ v_4)^T$ voltage.
$i = (i_1,\ i_2,\ i_3)^T$ current.
s : current applied to each node.

$$C = \begin{pmatrix} C_1 & 0 & 0 & 0 \\ 0 & C_2 & 0 & 0 \\ 0 & 0 & C_3 & 0 \\ 0 & 0 & 0 & C_4 \end{pmatrix}, L = \begin{pmatrix} L_1 & 0 & 0 \\ 0 & L_2 & 0 \\ 0 & 0 & L_3 \end{pmatrix}$$

Graph wave equation

$$C\frac{\mathrm{d}^2 v}{\mathrm{d}t^2} + \nabla^T L^{-1} \nabla\ v = \frac{\mathrm{d}s}{\mathrm{d}t}. \tag{10}$$

Network wave equation

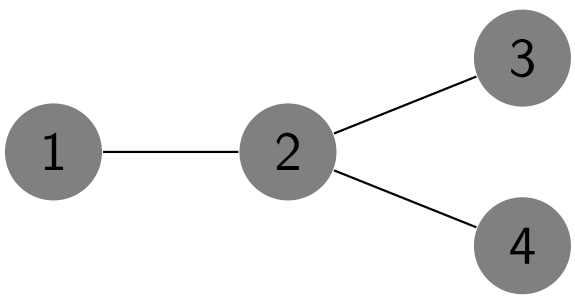
$$C\frac{\mathrm{d}^2 v}{\mathrm{d}t^2} + \nabla^T L^{-1} \nabla \; v = \frac{\mathrm{d}s}{\mathrm{d}t}. \tag{11}$$

$$\nabla^T L^{-1} \nabla = \begin{pmatrix} L_1^{-1} & -L_1^{-1} & 0 & 0 \\ -L_1^{-1} & L_1^{-1} + L_2^{-1} + L_3^{-1} & -L_2^{-1} & -L_3^{-1} \\ 0 & -L_2^{-1} & L_2^{-1} & 0 \\ 0 & -L_3^{-1} & 0 & L_3^{-1} \end{pmatrix}. \tag{12}$$

$$\frac{d^2 v}{dt^2} = -\Delta v. \tag{13}$$

Graph Laplacian :

$$\Delta = \nabla^T \nabla. \tag{14}$$



Paw graph.

$$\Delta = \begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 3 & -1 & -1 \\ 0 & -1 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{pmatrix}$$

Graph Laplacian

Adjacency matrix of $\mathcal{G}(\mathcal{V}, \mathcal{E})$:

$$A_{ij} = \begin{cases} 1 & \text{iff} \quad ij \in \mathcal{E}(\mathcal{G}) \quad (i \sim j) \\ 0 & \text{otherwise} \end{cases} \tag{15}$$

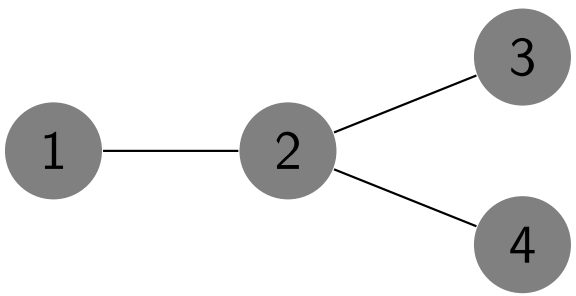
Vertices degree matrix:

$$D_{ii} = d_i = \sum_{j=1}^{N} A_{ij}. \tag{16}$$

Laplacian matrix:

$$\Delta = \nabla^T \nabla = D - A. \tag{17}$$

Example:



Paw graph.

$$\Delta = \begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 3 & -1 & -1 \\ 0 & -1 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{pmatrix}$$

Spectrum of graph Laplacian

Laplacian matrix Δ: real, symmetric

$$\Delta\ v^j = \omega_j^2\ v^j, \quad j \in \{1, \ldots, N\}\,. \tag{18}$$

- Eigenvalues $\omega_1^2 = 0 < \omega_2^2 \leq \cdots \leq \omega_N^2$.
- Eigenvectors v^j : orthonormal.
- $\omega_1 = 0 \longrightarrow$ Goldstone mode $v^1 = \frac{1}{\sqrt{N}}\,(1, 1, \ldots, 1)^T$.
- Multiplicity of the eigenvalue 0 $=$ number of connected components of $\mathcal{G}$.

Normal modes

$$\frac{d^2u}{dt^2} = -\Delta u. \tag{19}$$

$$\Delta\ v^j = \omega_j^2\ v^j, \quad j \in \{1, \ldots, N\}\,. \tag{20}$$

$$u(t) = \sum_{j=1}^{N} a_j(t) v^j. \tag{21}$$

$$\frac{d^2a_j}{dt^2} = -\omega_j^2 a_j. \tag{22}$$

Linear normal modes (linear periodic orbits)

$$u(t) = \left[\frac{\mathrm{d}a_1}{\mathrm{d}t}(0)\ t + a_1(0)\right] v^1 + \sum_{j=2}^{N} \left[a_j(0)\cos(\omega_j t) + \frac{\mathrm{d}a_j}{\mathrm{d}t}(0)\ \frac{\sin(\omega_j t)}{\omega_j}\right] v^j. \tag{23}$$

The graph wave equation 4: forcing and damping

Amplitude equations

$$Y_{tt} = -\Delta Y + N(Y)$$

$$Y(t) = a_1(t)v^1 + a_2(t)v^2 + \cdots + a_n(t)v^n.$$

$$a_{\mathbf{1}\,tt} + \omega_{\mathbf{1}}^{\mathbf{2}} a_{\mathbf{1}} =< N(Y)|v^{\mathbf{1}} >,$$
$$a_{\mathbf{2}\,tt} + \omega_{\mathbf{2}}^{\mathbf{2}} a_{\mathbf{2}} =< N(Y)|v^{\mathbf{2}} >,$$
$$\cdots$$
$$a_{n\,tt} + \omega_{n}^{\mathbf{2}} a_{n} =< N(Y)|v^{n} >.$$

Forcing at node n_f, damping at node n_d

$$Y_{tt} = -\Delta Y + D(-dY_t) + Ff1$$

$$D(n,n) = 0,\ F(n,n) = 0 \ \text{ except } \ D(n_d,n_d) = 1,\ F(n_f,n_f) = 1$$

$$1 = (1,1,\ldots,1)^T.$$

$$\ddot{a}_j = -\omega_j^2 a_j - dv_{n_d}^j \sum_{k=1}^n v_{n_d}^k \dot{a}_k + fv_{n_f}^j, \quad j = 1,n$$

The graph wave equation 5: soft nodes

$v^j_{n_d,n_f} = 0, \quad \rightarrow$ **ineffective forcing or damping for** a_j

Control system $\frac{dY}{dt} + \Delta Y + BU = 0$
cannot be controlled on a soft node (Kalman)

Observed system $\frac{dY}{dt} + \Delta Y = 0, \quad Z = CY$
System cannot be observed on a soft node (Kalman)